

Arithmetic of a Self-Iterating Generating Function

**Proofs of the A396807 congruences, rational prime-power lifts,
and an exact period of 46,860 for the last two digits**

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Abstract

Let $A = x + O(x^2)$ be the formal power series defined by $A = x + A^{(5)}A^{(6)}$, where angle brackets denote functional iteration. We prove the two congruences recorded as conjectures in OEIS A396807: $[x^n]A \equiv 1 \pmod{10}$ and $[x^n]A^{(k)} \equiv k^{n-1} \pmod{10}$ for every integer iterate k and every $n \geq 1$. The key is uniqueness over arbitrary coefficient rings and the substitution $\mathcal{M}(x) = x/(1-x)$.

The main extension is an arithmetic theory of the entire family $F = x + F^{(r)}F^{(s)}$. Its largest uniform Möbius-congruence modulus is exactly $\gcd(r+s-1, rs)$. For each prime dividing this modulus, we explicitly invert the linearized functional equation using a cyclic trace. Consequently, every prime-power reduction is rational and its coefficient sequence is eventually periodic. For A396807 we calculate the reductions modulo 4 and 25 and prove that the last two decimal digits have minimal period 46860, starting at $n = 1$. We also determine exact compositional orders, construct 2-adic and 5-adic iterates, prove zero complex radius of convergence, and settle closely related congruences in two neighboring OEIS entries. All proofs are accompanied by exact, reproducible computation; numerical tests are not used as a substitute for the all-degree arguments.

Status and scope. The OEIS records retrieved on September 19, 2026 still label the target statements as conjectures. This report supplies self-contained proofs and further deductions. A limited search did not locate a separate proof of the target statements; this is not a claim of exhaustive literature coverage or priority. The report has not been externally peer reviewed or formalized in a proof assistant.

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1 The problem, the answer, and the notation

1.1 A concrete conjecture

Paul D. Hanna's OEIS entry A396807, contributed on June 16, 2026, defines the sequence $a(n)$ by the ordinary formal generating function

$$A(x) = \sum_{n \geq 1} a(n)x^n = x + A^{(5)}(x)A^{(6)}(x). \quad (1)$$

The entry explicitly uses the exponents for *iteration*, not ordinary multiplication, and records the two congruences addressed below [1, 2].

Throughout the report,

$$F^{(0)}(x) = x, \quad F^{(j+1)} = F \circ F^{(j)}.$$

For negative j , $F^{(j)}$ is an iterate of the compositional inverse. Such an inverse exists over any commutative ring for a series $F = x + O(x^2)$. Ordinary powers, for example $F(x)^2$, retain their usual multiplicative meaning. This distinction is essential: replacing the right side of (1) by $x + A(x)^{11}$ gives an entirely different problem.

The first coefficients, also independently reproduced by the supplied program, are

$$1, 1, 11, 201, 4721, 129671, 3976201, 132554541, 4724742051, 178052854981, \dots$$

All congruences between power series in this report are coefficientwise. They do not assert convergence at a numerical value of x .

Theorem 1.1 (The two target congruences). *The unique solution of (1) in $x + x^2\mathbb{Z}[[x]]$ satisfies*

$$A(x) \equiv \frac{x}{1-x} \pmod{10}, \quad A^{(k)}(x) \equiv \frac{x}{1-kx} \pmod{10} \quad (k \in \mathbb{Z}). \quad (2)$$

In particular, for $n \geq 1$,

$$a(n) \equiv 1 \pmod{10}, \quad [x^n]A^{(k)} \equiv k^{n-1} \pmod{10}.$$

At $k = 0, n = 1$, the last display means that the coefficient is 1; equivalently, use the convention $0^0 = 1$ in this one coefficient formula. The modulus 10 is the largest positive integer for which all the coefficients of A equal 1 modulo that integer.

The proof occupies Sections 2–3 and does not require computation. The stronger results are summarized here to make their logical status clear.

Result	Conclusion	Section
Exact modulus	A necessary and sufficient condition for the whole two-iterate family, including an integer weight	3
Rational lifting	Rational reduction modulo every p^e at each prime selected by the exact modulus	5
Decimal digits	Explicit formulas modulo 4, 20, 25, and 100; minimal two-digit period 46860	6
Iteration	Exact order p^e modulo p^e and well-defined p -adic-time iterates	7
Related entries	Modulo-6 results for A396797; all modulo-8 iterate formulas for A396798, with a corrected starting index	8
Growth	A factorial lower bound along each parity and zero complex radius of convergence	9

2 Formal existence, uniqueness, and reduction

2.1 A coefficientwise contraction

Let R be a commutative ring with identity. Congruence modulo x^N means equality of the coefficients in degrees less than N . The x -adic topology simply packages this coefficientwise stabilization. For fixed nonnegative integers r, s and $c \in R$, set

$$T_c(F) = x + c F^{(r)} F^{(s)}, \quad F \in x + x^2 R[[x]]. \quad (3)$$

Lemma 2.1 (One extra coefficient). *If $F \equiv G \pmod{x^N}$ with $N \geq 2$, then*

$$F^{(j)} \equiv G^{(j)} \pmod{x^N} \quad (j \geq 0), \quad T_c(F) \equiv T_c(G) \pmod{x^{N+1}}.$$

Proof. Composition by a series of order one preserves divisibility by x^N . Also, if $U - V$ is divisible by x^N and $U(0) = V(0) = 0$, then $H(U) - H(V)$ is divisible by x^N : apply the identity that $U^m - V^m$ is divisible by $U - V$ to each coefficient of H . Induction on j , using

$$F \circ F^{(j)} - G \circ G^{(j)} = (F - G) \circ F^{(j)} + G \circ F^{(j)} - G \circ G^{(j)},$$

proves the assertion about iterates. The difference of the two products in (3) is

$$(F^{(r)} - G^{(r)})F^{(s)} + G^{(r)}(F^{(s)} - G^{(s)}).$$

Every iterate on the right has zero constant term, including the zeroth iterate x . Each summand is therefore divisible by x^{N+1} . No division, cancellation of ring elements, or field hypothesis is used. $\square$

Theorem 2.2 (Well-posedness over arbitrary rings). *There is a unique $F \in x + x^2 R[[x]]$ satisfying*

$$F = x + c F^{(r)} F^{(s)}. \quad (4)$$

Its coefficient of x^2 is c . The construction commutes with every homomorphism of coefficient rings.

Proof. Starting at $F_0 = x$, put $F_{j+1} = T_c(F_j)$. The first difference is cx^2 , and Lemma 2.1 shows inductively that $F_{j+1} - F_j$ is divisible by x^{j+2} . Every coefficient stabilizes, defining a series F . Each fixed-degree coefficient of a composition or product depends on only finitely many coefficients of its inputs. Passing to the stabilized coefficients gives $F = T_c(F)$.

Any two solutions agree modulo x^2 . Repeated application of the lemma makes them agree modulo every power of x , hence they are equal. The product in (4) begins with x^2 , proving the claim about its coefficient. Ring homomorphisms preserve all the finite sums and products used to construct a fixed coefficient, so construction and coefficient reduction commute. $\square$

2.2 Why negative iterates cause no difficulty

If $F = x + \sum_{n \geq 2} u_n x^n$, the coefficient of x^n in $F(G)$, where $G = x + \sum_{n \geq 2} v_n x^n$, is v_n plus a polynomial with integer coefficients in previously determined v_j and the u_j . Setting $F(G) = x$ therefore determines each v_n uniquely without dividing by a nonunit. The same argument constructs a left inverse; associativity of composition makes the left and right inverses equal.

It follows that inversion, positive iteration, and negative iteration all commute with reduction modulo an integer. In particular, the uniqueness theorem may be used directly over $\mathbb{Z}/10\mathbb{Z}$, although this ring is not a field. This is the exact logical step that promotes a rational substitution from an observation to a proof for every coefficient.

3 The Möbius substitution and the exact modulus

3.1 A calculation valid in every ring

For c in a coefficient ring define

$$M_c(x) = \frac{x}{1 - cx}.$$

The denominator has constant term 1, so the expression is a formal power series over the same ring. Direct composition gives

$$M_u \circ M_v = M_{u+v}, \quad M_c^{(j)}(x) = \frac{x}{1 - jcx} \quad (j \in \mathbb{Z}). \quad (5)$$

Write $S = r + s$ and $P = rs$. Clearing only denominators that are formal units gives the identity

$$M_c - T_c(M_c) = \frac{-c^2(S-1)x^3 + c^3Px^4}{(1-cx)(1-rsx)(1-scx)}. \quad (6)$$

Thus the two numbers $c^2(S-1)$ and c^3P govern the entire coefficientwise congruence.

Theorem 3.1 (Sharp uniform modulus). *Let $r, s \geq 0$ and $c \in \mathbb{Z}$. Let $F_{r,s,c} = x + \sum_{n \geq 2} u_n x^n$ be the unique integral solution of (4). For every positive integer m ,*

$$F_{r,s,c} \equiv M_c \pmod{m} \iff m \mid c^2(r+s-1) \text{ and } m \mid c^3rs. \quad (7)$$

Equivalently,

$$\gcd_{n \geq 2} |u_n - c^{n-1}| = \gcd(c^2(r+s-1), c^3rs). \quad (8)$$

We use $\gcd(0, 0) = 0$; a zero right side means exact equality, and hence congruence for every modulus. For $c = 1$ the sharp modulus is

$$D_{r,s} = \gcd(r+s-1, rs). \quad (9)$$

Proof. If the two divisibilities hold, (6) shows that M_c is a solution of the defining equation over $\mathbb{Z}/m\mathbb{Z}$. By Theorem 2.2, it is the reduction of $F_{r,s,c}$.

For necessity, the first four coefficients suffice. The elementary iteration expansion is

$$F^{(j)} = x + jcx^2 + (ju_3 + j(j-1)c^2)x^3 + O(x^4).$$

Substitution in the defining equation gives

$$F_{r,s,c} = x + cx^2 + c^2Sx^3 + c^3(2S^2 - S - P)x^4 + O(x^5). \quad (10)$$

Consequently, its discrepancies from M_c in degrees 3 and 4 are

$$c^2(S-1), \quad c^3((S-1)(2S+1) - P).$$

If both vanish modulo m , subtracting an integer multiple of the first from the second yields $m \mid c^3P$. These same two discrepancies have $\gcd(c^2(S-1), c^3P)$. Since that $\gcd$ divides every discrepancy by the sufficient direction, (8) follows. $\square$

Corollary 3.2 (All integer iterates). *Whenever (7) holds,*

$$F_{r,s,c}^{(k)}(x) \equiv \frac{x}{1 - kcx} \pmod{m} \quad (k \in \mathbb{Z}).$$

For $n \geq 1$, the corresponding coefficient is $(kc)^{n-1}$, with the degree-one convention stated in Theorem 1.1.

Proof. Reduction commutes with iteration and inversion, and (5) gives the iterates of the reduced series. $\square$

3.2 The target conjecture and a useful subfamily

For $(r, s, c) = (5, 6, 1)$,

$$D_{5,6} = \gcd(10, 30) = 10.$$

More concretely,

$$(1 - 5x)(1 - 6x) = 1 - 11x + 30x^2 \equiv 1 - x \pmod{10},$$

so that

$$x + \frac{x}{1 - 5x} \frac{x}{1 - 6x} \equiv x + \frac{x^2}{1 - x} = \frac{x}{1 - x} \pmod{10}.$$

Uniqueness proves the first target congruence, and Corollary 3.2 proves the second. Finally $a(3) - 1 = 10$, so a larger common modulus is impossible. This proves Theorem 1.1.

For adjacent iterate indices $(r, s) = (q, q + 1)$ with $q \geq 1$,

$$D_{q,q+1} = \gcd(2q, q(q + 1)) = \begin{cases} 2q, & q \text{ odd,} \\ q, & q \text{ even.} \end{cases} \quad (11)$$

The modulus is therefore not uniformly $2q$: the parity of q matters. This explains simultaneously the moduli 6, 4, and 10 in the three examples studied here.

Remark 3.3 (Scaling). If $F_{r,s,1} = x + \sum_{n \geq 2} a_n x^n$, then

$$F_{r,s,c}(x) = x + \sum_{n \geq 2} c^{n-1} a_n x^n.$$

For nonzero c , this is the identity $F_{r,s,c}(x) = c^{-1} F_{r,s,1}(cx)$, interpreted coefficientwise; it also holds at $c = 0$ without any division. Direct substitution and uniqueness prove the assertion.

4 A cyclic-trace inverse for the linearized equation

The uniform congruence is elementary. To go beyond it, we must control how the self-iteration equation changes under a small perturbation. The crucial fact is that the reduced Möbius map has a finite orbit. This section gives an explicit inverse, rather than invoking an unspecified lifting theorem.

4.1 The cyclic substitution and its trace

Fix a prime $p \mid D_{r,s}$. In $\mathbb{F}_p$, the conditions $r + s = 1$ and $rs = 0$ force the residues of r, s to be 0, 1 in some order. Interchanging them does not change the functional equation. We may therefore write

$$r = p\alpha, \quad s = 1 + p\beta, \quad \alpha, \beta \in \mathbb{Z}_{\geq 0}. \quad (12)$$

In formulas over $\mathbb{F}_p$, α and β mean their residues. Put

$$M_j(x) = \frac{x}{1 - jx}, \quad M = M_1, \quad M_0 = x.$$

Equation (5) implies $M_p = x$ over $\mathbb{F}_p$. For a formal series or a rational function regular at zero define

$$\mathrm{Tr}_p f = \sum_{j=0}^{p-1} f(M_j(x)). \quad (13)$$

The trace is invariant under $x \mapsto M(x)$, by cyclically reindexing the sum. If B is invariant, then $\mathrm{Tr}_p(Bf) = B \mathrm{Tr}_p(f)$.

Lemma 4.1 (Trace of the coordinate). *Over $\mathbb{F}_p(x)$,*

$$\mathrm{Tr}_p x = -\frac{x^p}{1 - x^{p-1}}. \quad (14)$$

Consequently, with

$$C(x) = \alpha x + \beta M(x), \quad (15)$$

we have

$$\mathrm{Tr}_p C = -\frac{(\alpha + \beta)x^p}{1 - x^{p-1}}. \quad (16)$$

Proof. In $\mathbb{F}_p[t]$, $\prod_{j \in \mathbb{F}_p} (t - j) = t^p - t$. Taking its logarithmic derivative gives

$$\sum_{j \in \mathbb{F}_p} \frac{1}{t - j} = -\frac{1}{t^p - t}.$$

Substituting $t = 1/x$ proves (14). The trace of $M(x)$ equals that of x , again by cyclic reindexing, proving (16). $\square$

4.2 First variation of a functional iterate

Work temporarily over $\mathbb{F}_p[\varepsilon]/(\varepsilon^2)$. Write a perturbation in the normalized form

$$F_\varepsilon = M + \varepsilon M^2 w.$$

The factor M^2 simplifies the chain rule.

Lemma 4.2 (Iteration derivative). *For every $k \geq 0$,*

$$F_\varepsilon^{(k)} = M_k + \varepsilon M_k^2 \sum_{j=0}^{k-1} w(M_j(x)). \quad (17)$$

The sum is empty at $k = 0$.

Proof. The formula is immediate at $k = 0, 1$. In passing from k to $k + 1$, the perturbation already present in the inner iterate is multiplied by $M'(M_k)$. Since

$$M'(M_k)M_k^2 = M_{k+1}^2,$$

and the new outer perturbation contributes $M_{k+1}^2 w(M_k)$, the partial sum acquires exactly one further term. $\square$

Let $T(F) = x + F^{(r)} F^{(s)}$ and denote by

$$Lh = h - (DT)_M h$$

the linearization of $F - T(F)$ at M . In the notation (12), the sums in (17) are

$$\sum_{j=0}^{r-1} w(M_j) = \alpha \mathrm{Tr}_p w, \quad \sum_{j=0}^{s-1} w(M_j) = \beta \mathrm{Tr}_p w + w.$$

Using $M_r = x$ and $M_s = M$ in $\mathbb{F}_p[[x]]$ gives the central identity

$$L(M^2 w) = M^2(1 - x)(w - C \mathrm{Tr}_p w). \quad (18)$$

For clarity, the two product-derivative terms, divided by M^2 , are $x(1 - x)\alpha \mathrm{Tr}_p w$ and $x(\beta \mathrm{Tr}_p w + w)$; subtracting them from w yields (18).

Theorem 4.3 (Explicit inverse). *For any $R \in x^3\mathbb{F}_p[[x]]$, set*

$$d = \frac{R}{M^2(1-x)}, \quad w = d + C \frac{\text{Tr}_p d}{1 - \text{Tr}_p C}, \quad h = M^2 w. \quad (19)$$

Then $h \in x^3\mathbb{F}_p[[x]]$ is the unique solution of $Lh = R$. If R is rational and regular at zero, then so is h .

Proof. The equation $Lh = R$ is equivalent to $w - C \text{Tr}_p w = d$. Taking the trace and using the invariance of $\text{Tr}_p w$ gives

$$(1 - \text{Tr}_p C) \text{Tr}_p w = \text{Tr}_p d.$$

The factor $1 - \text{Tr}_p C$ has constant term 1, so it is invertible. Substituting the resulting trace into the original equation gives (19), and proves uniqueness as well.

Since R starts in degree at least 3, d starts in degree at least 1. Both C and $\text{Tr}_p d$ have zero constant term. Thus w starts in degree at least 1 and $M^2 w$ in degree at least 3. The operations in (19) are finite rational operations and substitution by the rational functions M_j , all regular at zero. They preserve rationality and regularity. $\square$

5 Rational reduction modulo every selected prime power

5.1 The lifting theorem

Here a series over $\mathbb{Z}/p^e\mathbb{Z}$ is called rational when it can be written as $P(x)/Q(x)$ with polynomial numerator and denominator and with $Q(0)$ a unit. This definition is meaningful even though the coefficient ring is not a field.

Theorem 5.1 (Rational prime-power lifts). *Let $F = F_{r,s,1}$ and let p be a prime dividing $D_{r,s}$. For every $e \geq 1$, the coefficientwise reduction of F modulo p^e is rational. The proof gives a finite recursive construction of a rational representative.*

Proof. Let $\mathbb{Z}_{(p)}$ denote the rational numbers whose denominators are not divisible by p . Work with rational functions having coefficients in $\mathbb{Z}_{(p)}$ and a denominator whose constant term is a p -adic unit. Their expansions at zero lie in $\mathbb{Z}_{(p)}[[x]]$.

The initial representative is $f_1 = M$, by Theorem 3.1. Inductively, suppose that the rational function f_e agrees with F modulo p^e and has its coefficients of x and x^2 exactly equal to 1, with zero constant term. Its residual is rational and satisfies

$$T(f_e) - f_e \equiv 0 \pmod{p^e}.$$

The quotient

$$R_e = \frac{T(f_e) - f_e}{p^e} \pmod{p} \quad (20)$$

is a rational element of $x^3\mathbb{F}_p[[x]]$. To justify the division algebraically, write the residual over a common denominator with unit constant term. Multiplication by this denominator shows that every coefficient of its polynomial numerator is divisible by p^e in $\mathbb{Z}_{(p)}$. Dividing the numerator by p^e preserves the permitted class of rational functions.

Solve $Lh_e = R_e$ by Theorem 4.3. Choose a rational lift $\tilde{h}_e \in x^3\mathbb{Z}_{(p)}[[x]]$ of h_e , with denominator still having unit constant term, and put

$$f_{e+1} = f_e + p^e \tilde{h}_e. \quad (21)$$

Because $2e \geq e + 1$, terms involving two perturbations vanish modulo p^{e+1} . Also $f_e \equiv M \pmod{p}$, so its derivative operator reduces to $(DT)_M$. Hence

$$T(f_{e+1}) - f_{e+1} \equiv p^e (R_e + (DT)_M h_e - h_e) = 0 \pmod{p^{e+1}}.$$

Uniqueness over $\mathbb{Z}/p^{e+1}\mathbb{Z}$ identifies f_{e+1} with the reduction of F . It is rational and still has exactly the prescribed first two coefficients. This completes the induction. $\square$

Corollary 5.2 (Eventual periodicity). *For every such p and every $e \geq 1$, the coefficient sequence of F is eventually periodic modulo p^e . In particular, A396807 is rational modulo 10^e , and its last e decimal digits are eventually periodic, for every $e \geq 1$.*

Proof. Normalize a rational denominator to have constant term 1. Its coefficients impose a fixed finite-order forward linear recurrence after the degree of the numerator. Over a finite ring there are only finitely many states consisting of the last fixed number of coefficients. A repeated state gives a repeated future, proving eventual periodicity.

For A396807 use $p = 2, 5$. Rational representatives modulo 2^e and 5^e can have their denominators lifted with constant term exactly 1. A Chinese-remainder linear combination is then a rational representative modulo 10^e . The same finite-state argument applies. $\square$

Remark 5.3 (What the theorem does not say). The construction proves rationality separately at each finite modulus. It does not assert that one rational function over $\mathbb{Q}$ works at every modulus, nor that its numerator and denominator degrees stay bounded as e increases. It supplies no small uniform bound on the periods at higher powers. In fact, Section 9 proves that the characteristic-zero generating function is not rational.

5.2 A closed formula for the first new digit

The first lifting step is particularly simple because its normalized residual is exactly C .

Theorem 5.4 (Universal first correction). *With $r = p\alpha$ and $s = 1 + p\beta$, define in $\mathbb{F}_p(x)$*

$$H_{p;\alpha,\beta}(x) = \frac{x^3(\alpha(1-x) + \beta)(1-x^{p-1})}{(1-x)^3(1-x^{p-1} + (\alpha + \beta)x^p)}. \quad (22)$$

Then

$$F_{r,s,1}(x) \equiv M(x) + pH_{p;\alpha,\beta}(x) \pmod{p^2}. \quad (23)$$

On the right, any integral rational lift of H with unit denominator constant may be used.

Proof. Identity (6), with $c = 1$ and with its sign reversed, gives

$$\frac{T(M) - M}{p} \equiv \frac{x^3(\alpha + \beta - \alpha x)}{(1-x)^2} \pmod{p}.$$

Indeed, $(r + s - 1)/p = \alpha + \beta$ and $rs/p = \alpha(1 + p\beta) \equiv \alpha \pmod{p}$. After division by $M^2(1-x)$, this residual becomes

$$d = \alpha x + \beta \frac{x}{1-x} = C.$$

Formula (19) therefore simplifies to $w = C/(1 - \text{Tr}_p C)$. Multiply by M^2 and apply (16) to obtain (22). The lifting proof then gives (23). $\square$

This formula also explains why Artin–Schreier polynomials appear in a question that initially seems to concern only the last decimal digit. Writing $\lambda = \alpha + \beta$, the reciprocal of the new denominator factor is

$$t^p - t + \lambda.$$

If $\lambda \neq 0$ in $\mathbb{F}_p$, any root θ satisfies $\theta^{p^j} = \theta - j\lambda$. Its Frobenius orbit therefore has length exactly p , proving that this polynomial is irreducible of degree p . This elementary observation will be decisive for the exact period.

6 Explicit refinements and the two-digit period

6.1 Reductions modulo 4 and 25

For A396807 the applicable orientations in (12) are

p	r	s	$(\alpha, \beta) \bmod p$
2	6	5	(1, 0)
5	5	6	(1, 1).

Substitution into Theorem 5.4 yields the following particularly convenient representatives:

$$H_2(x) = \frac{x^3}{1 + x^3}, \quad (24)$$

$$\begin{aligned} H_5(x) &= \frac{x^3(2-x)(1-x^4)}{(1-x)^3(1-x^4+2x^5)} \\ &= \frac{x^3(2-x)(1+x+x^2+x^3)}{(1-x)^2(1-x^4+2x^5)}. \end{aligned} \quad (25)$$

The simplification of H_2 is performed over $\mathbb{F}_2$; the displayed rational function is then an integral lift of that result.

Corollary 6.1 (Refinements of the target congruence). *Coefficientwise,*

$$A \equiv M + 2H_2 \pmod{4}, \quad (26)$$

$$A \equiv M + 5H_5 \pmod{25}, \quad (27)$$

$$A \equiv M + 10 \frac{x^3}{1-x^3} \pmod{20}, \quad (28)$$

$$A \equiv M + 50H_2 + 80H_5 \pmod{100}. \quad (29)$$

In particular,

$$a(n) \equiv \begin{cases} 11 \pmod{20}, & 3 \mid n, \\ 1 \pmod{20}, & 3 \nmid n, \end{cases} \quad n \geq 1. \quad (30)$$

Proof. The first two statements are the two first-correction formulas. Modulo 4, $2x^3/(1+x^3) = 2x^3/(1-x^3)$, while modulo 5 the correction is zero. The Chinese remainder theorem proves (28). For (29), note that $50 \equiv 2 \pmod{4}$, $50 \equiv 0 \pmod{25}$, $80 \equiv 0 \pmod{4}$, and $80 \equiv 5 \pmod{25}$. $\square$

The first few last-two-digit values are a useful check, but not a proof of periodicity:

n	1	2	3	4	5	6	7	8	9
$a(n) \bmod 100$	01	01	11	01	21	71	01	41	51
n	10	11	12	13	14	15	16	17	18
$a(n) \bmod 100$	81	81	31	81	61	71	41	61	11

6.2 A denominator criterion for a minimal period

Lemma 6.2. *Let $H = N/Q \in \mathbb{F}_p(x)$ be in reduced form, with $Q(0) \neq 0$ and $\deg N \leq \deg Q$. The coefficients $h_n = [x^n]H$, indexed by $n \geq 1$, have period L if and only if Q divides $1 - x^L$.*

Proof. If $h_{n+L} = h_n$ for $n \geq 1$, then

$$H = h_0 + \frac{\sum_{j=1}^L h_j x^j}{1 - x^L},$$

so the reduced denominator divides $1 - x^L$. Conversely, if it divides, $(1 - x^L)H$ is a polynomial of degree at most L . Comparing coefficients above degree L gives exactly $h_{n+L} = h_n$ for $n \geq 1$. $\square$

Theorem 6.3 (Exact two-digit period). *The sequence $(a(n) \bmod 100)_{n \geq 1}$ is periodic from its first term, with minimal period*

$$\boxed{46860}. \quad (31)$$

The minimal periods modulo 4 and modulo 25 are 3 and 15620, respectively.

Proof. Modulo 4, formula (26) gives the repeating pattern 1, 1, 3, hence minimal period 3.

Modulo 25, the period equals the period of the coefficients of H_5 over $\mathbb{F}_5$, since $a(n) = 1 + 5[x^n]H_5$ in $\mathbb{Z}/25\mathbb{Z}$ for $n \geq 1$. In (25) its reduced denominator is

$$Q(x) = (1 - x)^2 Q_5(x), \quad Q_5(x) = 1 - x^4 + 2x^5. \quad (32)$$

To check that it is reduced, first evaluate the numerator at $x = 1$: it is $4 \neq 0$ in $\mathbb{F}_5$. Next, the reciprocal polynomial

$$q(t) = t^5 - t + 2$$

is irreducible. Indeed, if $q(\theta) = 0$, then $\theta^{5j} = \theta - 2j$, whose first return to θ occurs at $j = 5$. Thus the Frobenius orbit, and hence the minimal polynomial, has degree 5. The numerator of (25) is a product of x^3 , a linear factor, and a degree-three factor, none of which is divisible by this irreducible reciprocal factor.

Let θ be the class of t in $\mathbb{F}_5[t]/(q)$. The multiplicative order of θ divides $5^5 - 1 = 3124 = 4 \cdot 11 \cdot 71$. Reduction of powers using $\theta^5 = \theta - 2$ gives

$$\theta^{1562} = 4, \quad (33)$$

$$\theta^{284} = 4 + \theta + 2\theta^2 + 4\theta^4, \quad (34)$$

$$\theta^{44} = 2 + 2\theta + \theta^2 + 3\theta^3 + 3\theta^4. \quad (35)$$

None is 1, because a nonzero polynomial of degree less than 5 cannot vanish at θ . The three exponents are $3124/2$, $3124/11$, and $3124/71$; they exclude every possible proper divisor as the order. Hence θ , and also its reciprocal, has order exactly 3124. The displayed residues are small, independently checkable certificates; the supplied program obtains them by binary powering and reduction.

It follows that Q_5 divides $1 - x^L$ precisely when $3124 \mid L$. The factor $(1 - x)^2$ divides $1 - x^L$ precisely when $5 \mid L$: write $L = 5^v u$ with $5 \nmid u$ and use $1 - x^L = (1 - x^u)^{5^v}$. By Lemma 6.2, the minimal period of H_5 is therefore

$$\text{lcm}(5, 3124) = 15620.$$

The degree condition in that lemma is satisfied: both the numerator and denominator in the reduced expression for H_5 have degree 7. Thus there is no initial transient for the coefficients indexed from $n = 1$.

A period modulo 100 must be a period modulo both 4 and 25; conversely, a common period works by the Chinese remainder theorem. The minimal period is consequently $\text{lcm}(3, 15620) = 46860$. $\square$

6.3 The general first-lift period mechanism

The same proof gives more than this numerical example.

Proposition 6.4. *Assume $\lambda = \alpha + \beta \neq 0$ in $\mathbb{F}_p$ and let ℓ be the multiplicative order of a root of $t^p - t + \lambda$ in $\mathbb{F}_{p^p}$. The coefficients of $F_{r,s,1}$ modulo p^2 , indexed from $n = 1$, have minimal period*

$$\begin{cases} \ell, & \beta = 0 \text{ in } \mathbb{F}_p, \\ p\ell, & \beta \neq 0 \text{ in } \mathbb{F}_p. \end{cases}$$

In particular, ℓ divides $p^p - 1$. It need not be assumed equal to $p^p - 1$; that equality in the A396807 case was proved by the certificates above.

Proof. The Artin–Schreier factor is irreducible of degree p by the Frobenius-orbit calculation following Theorem 5.4. Cancel $1 - x$ from $1 - x^{p-1}$ in (22). The remaining numerator factors have degrees less than p , so none cancels the irreducible factor. At $x = 1$ their value is a nonzero multiple of β . When $\beta \neq 0$, the reduced denominator retains $(1 - x)^2$. When $\beta = 0$, $\alpha \neq 0$ and one further factor $1 - x$ cancels, leaving just a simple factor $1 - x$.

The degree of the reduced numerator is at most that of its denominator. Lemma 6.2 now requires $\ell \mid L$ and, only in the double-factor case, $p \mid L$. Since $\ell \mid p^p - 1$, it is prime to p . Formula (23) transfers exactly these periods from H over $\mathbb{F}_p$ to F modulo p^2 . $\square$

7 Exact compositional order and p-adic time

Periodicity in the coefficient index n must not be confused with periodicity in the iteration index k . The latter has a different, and particularly simple, exact answer.

Lemma 7.1 (Raising a near-identity map). *If $G(x) \equiv x \pmod{p^e}$ with $e \geq 1$, then $G^{\langle p \rangle}(x) \equiv x \pmod{p^{e+1}}$.*

Proof. Write $G = x + p^e h$. For any formal series h with zero constant term, $h(x + p^e h(x)) - h(x)$ has all coefficients divisible by p^e . Consequently composition of two near-identity maps adds their p^e -perturbations modulo p^{2e} . Since $2e \geq e + 1$, induction gives

$$G^{\langle j \rangle} \equiv x + jp^e h \pmod{p^{e+1}}.$$

Taking $j = p$ proves the claim. The argument includes $p = 2$ and $e = 1$. $\square$

Theorem 7.2 (Exact order). *For $F = F_{r,s,1}$ and any prime $p \mid D_{r,s}$, the order of the reduction of F in the composition group of $x + x^2(\mathbb{Z}/p^e\mathbb{Z})[[x]]$ is exactly p^e . In particular,*

$$A^{\langle 10^e \rangle}(x) \equiv x \pmod{10^e},$$

and the compositional order of A396807 modulo 10^e is exactly 10^e .

Proof. Modulo p , F is M , which has p th iterate x . Lemma 7.1 inductively gives $F^{\langle p^e \rangle} \equiv x \pmod{p^e}$. The coefficient of x^2 in F is 1, and the coefficient of x^2 in its k th iterate is k . Thus an iterate equal to x modulo p^e must have $p^e \mid k$, proving exactness. For A396807 combine the two prime-power orders by the Chinese remainder theorem, whose least common multiple is 10^e . $\square$

Corollary 7.3 (A well-defined p -adic iteration parameter). *For each selected prime p , there is an injective group homomorphism*

$$(\mathbb{Z}_p, +) \longrightarrow (x + x^2\mathbb{Z}_p[[x]], \circ), \quad t \longmapsto F^{\langle t \rangle},$$

which extends the usual integer iterates, and $[x^2]F^{\langle t \rangle} = t$.

Proof. Choose integers k_e representing t modulo p^e . Theorem 7.2 makes the reductions of $F^{\langle k_e \rangle}$ compatible as e increases, independently of the representatives. Take the inverse limit coefficient by coefficient. Every coefficient of a composition is a finite polynomial in input coefficients, so the integer identity $F^{\langle a \rangle} \circ F^{\langle b \rangle} = F^{\langle a+b \rangle}$ passes to the inverse limit. The degree-two coefficient is t , proving injectivity. $\square$

These are p -adic formal iterates, not real fractional iterates. No assertion about a real-time flow follows from the construction.

8 Two neighboring OEIS entries

8.1 A396797: the same proof modulo 6

The entry A396797 uses the equation $B = x + B^{\langle 3 \rangle} B^{\langle 4 \rangle}$ and records both the constant coefficient congruence and its all-integer-iterate version modulo 6 [3]. Here $D_{3,4} = 6$, so Theorem 3.1 and Corollary 3.2 immediately prove

$$B \equiv \frac{x}{1-x} \pmod{6}, \quad B^{\langle k \rangle} \equiv \frac{x}{1-kx} \pmod{6} \quad (k \in \mathbb{Z}).$$

Thus this is not a second unrelated computation: it is another specialization of the same exact-modulus theorem.

8.2 A396798: a corrected start and all its modulo-8 iterates

Let $G = F_{4,5,1}$, the generating function in A396798. The entry lists a repeating coefficient word modulo 8 and several patterns for its first eight iterates [4]. A single formula handles all of them, including a small indexing correction.

Theorem 8.1. *Set $J(x) = x^4/(1-x)^3$. For every integer k ,*

$$G^{\langle k \rangle}(x) \equiv \frac{x}{1-kx} + 4(k \bmod 2)J(x) \pmod{8}. \quad (36)$$

In particular,

$$G \equiv \frac{x}{1-x} + \frac{4x^4}{(1-x)^3} \pmod{8}.$$

The coefficients of G are 1 at $n = 1$, followed by the repeating word 1, 1, 5, 5 starting at $n = 2$. For odd k , the coefficients of $G^{(k)}$ starting at $n = 2$ repeat

$$k, 1, k+4, 5 \pmod{8}. \quad (37)$$

For even k , they are exactly the coefficients of $x/(1-kx)$ modulo 8.

Proof. The exact uniform modulus is $D_{4,5} = 4$, so $G \equiv M \pmod{4}$. The residual of M , divided by 4 and reduced modulo 2, is

$$\frac{T(M) - M}{4} = \frac{2x^3 - 5x^4}{(1-x)(1-4x)(1-5x)} \equiv \frac{x^4}{(1-x)^2} \pmod{2}.$$

For $(r, s, p) = (4, 5, 2)$, both $\alpha = 2$ and $\beta = 2$ vanish in $\mathbb{F}_2$, so $C = 0$ and (18) reduces to $Lh = (1-x)h$. The required correction is therefore $J = x^4/(1-x)^3$, proving $G \equiv M + 4J \pmod{8}$.

To iterate this identity, only the first variation modulo 2 matters: terms quadratic in $4J$ vanish modulo 8. Normalize $J = M^2w$, where

$$w = \frac{x^2}{1-x}.$$

A direct substitution in $\mathbb{F}_2(x)$ gives $w(M) = w(x)$. The sum in (17) is thus $(k \bmod 2)w$. For odd k , $M_k = M$ in $\mathbb{F}_2(x)$; for even k , the sum is zero. This proves (36) for nonnegative k . Theorem 7.2 gives $G^{(8)} = x$ modulo 8, and both sides of (36) depend only on k modulo 8, extending the formula to all integers.

For $n \geq 4$,

$$[x^n]J = \binom{n-2}{2};$$

in smaller positive degrees it is zero. Its parity, starting at $n = 2$, is the repeating word 0, 0, 1, 1. For odd k , k^{n-1} modulo 8 alternates $k, 1$ starting at $n = 2$. Adding the correction gives (37). $\square$

For instance, the even iterates are

$$G^{(2)} \equiv x + 2x^2 + 4x^3, \quad G^{(4)} \equiv x + 4x^2, \quad G^{(6)} \equiv x + 6x^2 + 4x^3, \quad G^{(8)} \equiv x \pmod{8}.$$

The odd words for $k = 3, 5, 7$ are respectively (3, 1, 7, 5), (5, 1, 1, 5), and (7, 1, 3, 5).

The literal proposed word for G cannot start at $n = 1$: its third coefficient is $9 \equiv 1 \pmod{8}$, not 5. The corrected start $n = 2$ gives an exact theorem. This is an indexing correction in the neighboring entry, not a counterexample to either of the two main A396807 conjectures.

9 Large integer coefficients and zero complex radius

The rational reductions do not reflect convergence of the original ordinary generating function. In fact its coefficients grow too quickly for any positive complex radius.

Theorem 9.1 (A factorial lower bound). *Let $F = F_{r,s,1} = \sum_{n \geq 1} a_n x^n$ with $\max(r, s) \geq 2$. Then all coefficients are nonnegative, and*

$$a_n \geq (n-2)a_{n-2} \quad (n \geq 3). \quad (38)$$

In particular, for $m \geq 1$,

$$a_{2m} \geq 2^{m-1}(m-1)!, \quad a_{2m+1} \geq (2m-1)!!. \quad (39)$$

The complex radius of convergence is zero.

Proof. The construction in Theorem 2.2, starting at x , uses only products, compositions, and addition with nonnegative coefficients. Hence the stabilized coefficients are nonnegative. Also $F \geq x$ coefficientwise and $a_1 = a_2 = 1$. Coefficientwise order is preserved by composition of such series, so any iterate of order at least two is at least $F \circ F$, and every other iterate is at least x . The defining equation thus implies

$$a_n \geq [x^{n-1}](F \circ F).$$

Within $F(F(x))$, select the outer term $a_{n-2}F(x)^{n-2}$. Because $F = x + x^2 + O(x^3)$, the coefficient of x^{n-1} in $F(x)^{n-2}$ is $n-2$: exactly one of its $n-2$ factors contributes x^2 and all others contribute x . This proves (38).

Iterating it along the two parities gives (39). These lower bounds have n th roots tending to infinity. For example, at least half the factors in $(m-1)!$ are at least $(m-1)/2$, which already makes the $2m$ th root unbounded. The odd case is similar. Therefore $\limsup_n |a_n|^{1/n} = \infty$, and the radius is zero. $\square$

This applies in particular to A396807. A rational function regular at zero has positive complex radius of convergence, so A is not rational over $\mathbb{Q}$. There is no contradiction with rationality modulo every 10^e : reduction discards unbounded amounts of integer information, and the rational representative is allowed to change with e . The lower bound is not an asymptotic equivalent. It also does not, by itself, decide whether the series is D-finite.

10 Algorithms, independent checks, and reproducibility

10.1 An exact triangular construction

Repeatedly recomputing the whole functional equation is unnecessary. The supplied `series_tools.py` maintains two tables:

$$P_{k,n} = [x^n]F(x)^k, \quad B_{j,n} = [x^n]F^{(j)}(x).$$

Here P uses ordinary powers and B uses functional iterates. Initially $P_{0,0} = 1$ and $B_{0,n}$ is 1 at $n = 1$ and 0 otherwise. At degree n , perform the following operations in order:

$$a_n = 1_{n=1} + c \sum_{i=1}^{n-1} B_{r,i} B_{s,n-i}, \quad (40)$$

$$P_{1,n} = a_n, \quad P_{k,n} = \sum_{i=1}^{n-k+1} a_i P_{k-1,n-i} \quad (2 \leq k \leq n), \quad (41)$$

$$B_{1,n} = a_n, \quad B_{j,n} = \sum_{k=1}^n B_{j-1,k} P_{k,n} \quad (2 \leq j \leq \max(r, s, 1)). \quad (42)$$

Equation (40) uses only previously completed degrees because both product factors have zero constant term. Equation (41) also uses only smaller degrees, and (42) is triangular in j . The last identity represents $F^{(j-1)} \circ F$, which equals the j th iterate. These observations prove the algorithm directly.

Up to degree N , it uses $O(N^3 + \max(r, s)N^2)$ ring operations and $O(N^2 + \max(r, s)N)$ ring elements of storage. These are arithmetic-operation bounds, not bit-complexity bounds; integer sizes grow substantially. Reducing after each operation gives the same algorithm over any desired finite modulus.

For the last two digits alone, (29) is much faster. Expanding a fixed rational function gives a fixed-order linear recurrence, hence $O(N)$ operations on bounded residues. After storing one period, any particular index is answered by reducing $n - 1$ modulo 46860 and adding one. This does not require constructing the huge integer $a(n)$.

10.2 What was actually checked

The standard-library verification program was run with Python 3.13.5. The symbolic certificates were independently run with SymPy 1.14.0. The generated files contain the full machine-readable logs.

Check	Executed range / result
Exact A396807 construction	$1 \leq n \leq 400$; $a(400)$ has 910 digits
Published initial values	First 19 values matched
Independent Horner-composition residual	Defining equation checked through degree 100
Main and refined congruences	2,000 coefficient checks through degree 400
Positive and negative iterate congruences	$-20 \leq k \leq 20$, $1 \leq n \leq 48$: 1,968 checks
Weighted sharp modulus and scaling	1,014 parameter triples; 29,406 coefficients
Compositional order upper bounds	Moduli 2, 4, 8, 16, 5, 25, 125, 625 through degree 40
Neighboring entries	100 A396797 checks; 580 A396798 checks
Period data	93,720 residues from the proved rational formulas; two full matching periods
Symbolic identities	Universal residual, both first corrections, direct chain-rule checks, trace checks, and the A396798 correction passed

The weighted family range was $0 \leq r, s \leq 12$, $c \in \{-3, -1, 0, 1, 2, 5\}$, through degree 30. For every triple, the gcd of the computed discrepancies equaled (8), and the scaling identity held coefficientwise.

The 400 large integer coefficients were independently generated; only the first 19 were compared with the published initial list. The 93,720 two-digit residues were obtained from the proved rational formulas, *not* by independently generating 93,720 full integer coefficients. This distinction prevents circular evidence: the all-degree rational identities and period proof are established algebraically above; the long residue file is an efficient consequence and an implementation check.

As additional implementation checks, the program rejects every maximal proper divisor of the proposed two-digit period. It finds the following discrepancies, all already at $n = 1$:

Proposed shift L	$a(1) \bmod 100$	$a(1 + L) \bmod 100$
23430	01	81
15620	01	51
9372	01	21
4260	01	61
660	01	81

These finite discrepancies are consistent with, but are not needed in place of, the denominator and finite-field proof of minimality.

10.3 Files and commands

The archive contains the LaTeX source and compiled PDF, plus:

File	Purpose
code/series_tools.py	Exact family solver, independent composition and inverse routines, rational recurrence expansion
code/verify.py	Standard-library verification and data generation
code/symbolic_checks.py	Optional SymPy rational identities and reusable first-correction/linearized-solver functions
data/a396807_exact_400.txt	Independently generated exact values
data/a396807_mod100_period.txt	One complete minimal two-digit period, with indices
data/verification.json	Detailed finite-test results
data/symbolic_verification.json	Symbolic certificate results
source_notes/status.md	Source provenance and status distinctions

From the unpacked directory, run

```
python code/verify.py
python code/symbolic_checks.py # optional; requires SymPy
latexmk -pdf -interaction=nonstopmode article.tex
```

The first command requires no third-party Python packages. The general higher-power rational lifting procedure is given explicitly in the proof of Theorem 5.1; the code implements its linearized inverse and closed first correction, not an optimized arbitrary-exponent rational-function engine.

11 Conclusions and further questions

The selected A396807 conjectures are proved, including negative functional iterates and sharpness of the modulus. The useful broader lesson is that reduction to a finite-order rational substitution can turn an equation involving unknown functional iterates into an explicitly invertible trace equation.

For this family the chain of implications is fully proved:

$$\begin{aligned}
 \text{sharp Möbius congruence} &\implies \text{cyclic-trace inverse} \\
 &\implies \text{rational prime-power lifts} \\
 &\implies \text{eventually periodic residues.}
 \end{aligned}$$

At the first additional digit the denominator is explicit enough to prove exact periods. For A396807 this produces a period of 46860 for the last two digits, despite superexponential lower bounds on the integer coefficients themselves.

Several questions remain beyond the results proved here. One is to obtain comparably explicit minimal periods modulo 10^e for $e \geq 3$, rather than merely eventual periodicity. Another is to control the growth of numerator and denominator degrees in the rational lifting algorithm. A third is to sharpen (39) to an actual asymptotic description of $a(n)$. Finally, the method as stated selects primes dividing $D_{r,s}$; it does not classify reductions at other primes. These are natural continuation questions, not claims here about their global status in the literature.

The computational evidence is deliberately separated from the proofs. The finite checks test implementations and catch algebraic mistakes; existence, congruence, rationality at every selected prime power, and minimal period are established by the preceding arguments.

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